University of Aberdeen
Designing Real World Trials - PU5068
Assessment 1 - Flow Diagram

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Rationale

The study wants to compare the effectiveness of long-term GnRH analogues with add-back HRT against laparoscopic surgery in terms of clinical and cost for managing recurrent pain in women with endometriosis.

1 PICOT

The PICOT of this study is as follows:

- P: Women aged between 21-49 years with pain recurrence following previous surgical treatment of endometriosis but wish to avoid removal of ovaries and hysterectomy
- I: Long term use of GnRH analogues with add-back HRT
- C: Laparoscopic ablation/excision of endometriosis
- O: Better functional health and well-being measured by EHP-30 and EQ-5D
- T: 54 months

2 Target population

The target population of this research is women aged between 21-49 years with pain recurrence following previous surgical treatment of endometriosis but wish to avoid removal of ovaries and hysterectomy. The main issue being addressed is endometriosis which is a very common condition in women of reproductive age with about 1 in 10 affected. Endometriosis significantly impacts quality of life because of the side effect it brings, such as pelvic pain. Currently, the standard approach is laparoscopic surgery followed with hormonal contraceptives if there are no plans on conceiving. However, this approach itself doesn't completely remove the chance of pain recurrence in the future and pain recurrence is fairly common with almost half the population being affected. Furthermore, the Scottish national data states that 45% of women diagnosed with endometriosis has had three or more surgeries. The target population was chosen because there is an increase in awareness on how the current standard laparoscopic surgery approach may not be as clinically and economically effective.

3 Eligibility of the numbers in the trial

The trial design plans to randomise 400 women to 2 groups: intervention (use of GnRH analogues with add-back HRT) and comparison (laparoscopic surgery) but the primary outcome data needs data on 320 women two years post randomisation. This is due to potential attrition that was set at 20% which can be verified by the following

$$400 - (400 \times 20\%) = 400 - 80 = 320$$

4 Informed consent and baseline measures

Clinical trial participation often carries potential risks for participants. While those in the intervention group receive the treatment that wants to be analysed, those in the control group may also receive some level of care which could be in the form of standard procedural care, both definitely has its own risks. So, it is highly important that the participants know and fully understand the risks involved in participating in the trial before giving their informed consent. Participants' informed consent should be obtained before any activity related to the trial is done so they understand the risks or benefits and enough time should be given to the participants before obtaining their consent.

Baseline measures act as the initial data collection before the intervention begins, it is important to know how the overall and individual measurement holds so that after treatment the effect of the intervention can be analysed by comparing a before and after measurement.

5 Exclusion criteria

Alongside inclusion criteria, exclusion criteria acts as the boundary and standard of the study population. It can reduce the risk of the intervention potentially having an adverse effect on the participants by acting as a safeguard. However, excision criteria (and inclusion criteria) need to be clearly defined and there exists a trade off between having too many of either criterion, too many exclusion can negatively impact the external validity.[1] This research has clearly defined the exclusion criterion so that there is a clear boundary of which participants are automatically not eligible.

6 Randomisation and treatment allocation

Randomisation protects against potential selection bias when assigning participants' groups by replacing the researcher preference with a randomly generated assignment. If the researcher manually assigns participants to the trial groups, they may subconciously assign the sickest participant to the intervention group to observe the most optimal results. Furthermore, randomisation should be coupled with allocation concealment by concealing which group the participants will be in from the participants themselves and the recruiter. Evidence suggests that trials that didn't have proper allocation concealment tend to overestimate the results of their findings.[2] Randomisation should always be done with allocation concealment to avoid any chance of selection bias. This trial mentions doing randomisation but does not mention how randomisation is done and no information on allocation concealment is provided.

7 Intervention and comparison

The intervention group in this trial is a medication approach in trying to treat endometriosis, while the comparison is the current standard procedure of la-

paroscopic surgery. Since there is a need for a better alternative than surgery that is non-surgical and with better results in pain recurrence, the intervention is adding low dose hormone replacement therapy (HRT) on top of GnRH analogues. Initially, GnRH analogues alone has been tested previously but on a shorter time period and they come with unwanted side effects like hot flushes and osteoporosis concerns. Low dose HRT has been shown to help reduce the side effects, thus there is a need to test whether it is effective long term.

8 Outcome measures and timing of outcome assessment

The primary outcome of this trial is measuring the participants' functional health and well being via the Endometriosis Health Profile-30 (EHP-30) and EQ-5D for their health related quality of life. The secondary clinical outcome measures and checks whether there is an adverse effect related to GnRH analogues that requires the treatment to be stopped as well as a DEXA scan to measure the participants bone mineral density. The primary outcome is taken 6, 12, 18, and 24 months after randomisation, while the secondary outcome is measured 12 and 24 months after randomisation.

9 Sample size accounting for dropout

The trial needs data from 320 women two years after randomisation to properly measure the primary outcome with a 90% power. However, with only 320 participants from the start of the trial to the end is likely impossible. Participant dropout is common in trials and needs to be accounted for, the trial mentions a 20% attrition rate which means that the total participants needed is 400 participants, the 320 participants for the primary outcome comes from the assumption that 20% of the recruited participants drop out in the middle of the trial.

References

1. Inclusion and Exclusion Criteria. Medsurg nursing. 2020; 29:125
2. Mitchell A, Moe-Byrne T, Cunningham-Burley R, Dean A, Rangan A, Roche J, and Torgerson DJ. Poor allocation concealment methods are associated with heterogeneity in age and statistical significance of the primary outcome: Review of recent trials published in four general medical journals. Journal of Evaluation in Clinical Practice 2020; 26:1316–9. DOI: https://doi.org/10.1111/jep.13313. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jep.13313. Available from: https://onlinelibrary.wiley.com/doi/abs/10.1111/jep.13313